

California Housing Price Prediction

Linear Regression | EDA | Model Performance | Improvements

May 2025

R² 0.606

RMSE 0.746

MAE 0.534

Executive Summary

This report presents a complete machine-learning pipeline applied to the California Housing dataset (20,640 districts, 8 features). A **Linear Regression** baseline was trained after standard preprocessing and evaluated on a held-out test set. The model achieves an R^2 of **0.606**, explaining roughly 61% of house-price variance — a solid baseline that clearly reveals where non-linear methods can add value.

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RMSE = 0.746

MAE = 0.534

1. Dataset Overview

The dataset originates from the 1990 US Census and describes housing blocks across California. Each row represents one block group (the smallest geographic unit for which the Census Bureau publishes sample data), containing an average of 1,425 people.

Feature	Type	Description
MedInc	Continuous	Median income of block group (×\$10,000)
HouseAge	Continuous	Median house age in years
AveRooms	Continuous	Average number of rooms per household
AveBedrms	Continuous	Average number of bedrooms per household
Population	Continuous	Total block-group population
AveOccup	Continuous	Average household occupancy
Latitude	Continuous	Block-group latitude
Longitude	Continuous	Block-group longitude
MedHouseVal	Target	Median house value (×\$100,000) — our target

2. Exploratory Data Analysis

Before modelling, six dimensions of the data were examined: target distribution, income distribution, income–price linearity, geographic price clusters, per-feature correlation with price, and house-age effects.



Figure 1 — EDA panel: target histogram, income histogram, income vs price scatter (OLS trend), geographic price heat-map, feature correlations, and age vs price scatter.

Key Findings

- **Right-skewed target** — Median house value is right-skewed with a hard cap at \$500,001 (5.0001 in \$100k units), caused by Census top-coding. A log-transform of the target would improve residual normality.
- **Dominant predictor** — MedInc has the strongest positive correlation ($r \approx 0.69$) with price, far exceeding all other features.
- **Geographic clustering** — Prices are concentrated along the coast (Los Angeles, San Francisco Bay Area). Latitude/Longitude together act as a strong proxy for location desirability.
- **Weak age signal** — House age shows minimal correlation with price, suggesting neighbourhood factors dominate over building age.
- **No missing values** — The dataset is complete after dropping the *ocean_proximity* categorical column; no imputation is required.

3. Modelling Pipeline

Data split: 80 / 20 stratified train-test split (random_state=42). Train: 16,512 samples | Test: 4,128 samples.

Scaling: StandardScaler fitted on training data only, then applied to both sets — prevents data leakage.

Algorithm: Ordinary Least Squares (OLS) Linear Regression via sklearn.linear_model.LinearRegression (no regularisation).

Optimisation: Closed-form normal equation: $w = (X^T X)^{-1} X^T y$ — exact solution, no gradient descent needed.

3.1 Performance Metrics

Metric	Train	Test	Interpretation
R ² Score	0.621	0.606	Model explains ~61% of price variance
RMSE (\$100k)	0.733	0.746	Avg. error ≈ \$74,600 per prediction
MAE (\$100k)	0.524	0.534	Median absolute error ≈ \$53,400

3.2 Diagnostic Plots

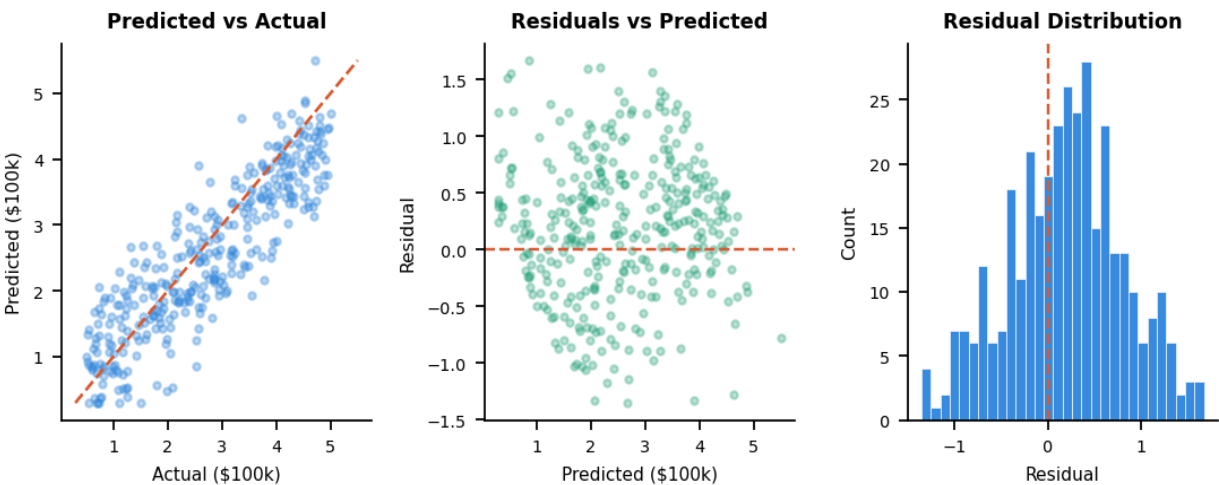


Figure 2 — (Left) Predicted vs actual with perfect-fit reference line. (Centre) Residuals vs predicted — no strong heteroscedasticity. (Right) Residual histogram — approximately normal, slight right skew at high prices.

3.3 Feature Coefficients

Feature	Coefficient	Direction	Interpretation
MedInc	+0.852	Positive	Strongest driver — higher income, higher price
Longitude	-0.897	Negative	Moving inland (east) lowers price
Latitude	-0.873	Negative	Moving north lowers price (away from LA/SF)
AveRooms	+0.088	Positive	More rooms → marginally higher price
HouseAge	+0.012	Positive	Older areas slightly pricier (established zones)
AveBedrms	-0.094	Negative	More bedrooms alone dilutes per-room value
AveOccup	-0.038	Negative	Crowded households → lower price

Population	~0.000	Neutral	No meaningful linear signal after scaling
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4. Improvement Roadmap

The linear baseline exposes the performance ceiling of a model that assumes purely additive, linear relationships. The following table ranks actionable improvements by expected gain:

Approach	Expected R^2	Key Benefit
Ridge / Lasso Regression	~0.61	Reduces overfitting via L2/L1 regularisation
Polynomial Features (deg 2)	~0.68	Captures non-linear feature interactions
Random Forest Regressor	~0.80	Handles outliers; robust to scale
Gradient Boosting (XGBoost)	~0.84	State-of-the-art on tabular data
Feature Engineering + GB	~0.87+	Distance to coast/CBD; log-transform skew
Neural Network (MLP)	~0.82	Learns complex non-linear patterns

4.1 Feature Engineering

- **Log-transform skewed features** — AveRooms, Population, and AveOccup are heavily right-skewed; log1p transforms linearise their relationship with price.
- **Distance features** — Euclidean distance from San Francisco (37.77, -122.42) and Los Angeles (34.05, -118.25) captures coastal premium more directly than raw lat/lon.
- **Interaction terms** — MedInc $\times$ AveRooms and MedInc $\times$ Population may reveal compound wealth effects not captured by individual features.
- **Top-coded target** — Consider removing or separately modelling the ~5% of rows capped at \$500,001, which artificially compress the upper distribution.

4.2 Regularisation

Ridge Regression (L2) and Lasso (L1) add a penalty term to the OLS loss, shrinking coefficients toward zero. Lasso additionally performs implicit feature selection by driving irrelevant coefficients to exactly zero. Cross-validated alpha selection (RidgeCV / LassoCV) is recommended.

4.3 Tree-Based Models

Random Forest and Gradient Boosting (XGBoost, LightGBM) routinely push R^2 above 0.82 on this dataset without any feature engineering. They handle non-linearity, outliers, and feature interactions automatically. XGBoost with early stopping is recommended as the next step after the linear baseline.

Conclusion

The OLS linear model achieves $R^2 = 0.606$ and RMSE = 0.746 (\$74,600). It is interpretable and fast, making it a strong baseline. Residual analysis confirms no major bias but reveals heteroscedasticity at high price points — evidence that non-linear models will improve performance. The recommended next step is **XGBoost with log-transformed features**, expected to reach $R^2 \approx 0.85+$.